

IMPROVING CLASSIFICATION WITH THE ADABOOST META-ALGORITHM

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PROBLEM STATEMENT

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- For a classification problem (assume binary), we are given a "weak classifier".
- Weak classifier - Classifier that performs just slightly better than random guessing ($> 50\%$ accuracy).
- Can we combine multiple instances of the weak classifier to obtain a strong classifier?

- Methods that combine multiple classifiers are called ensemble methods or meta-algorithms.
- Bagging and boosting are two common types.

BAGGING AND BOOSTING

- Given a dataset X , we randomly sample X (with replacement) S times to make S new datasets of equal size as X .
- The weak classifier is applied to each dataset individually.
- To classify a new data point, we apply our S classifiers to the new data points and take a majority vote.

- Sequential use of classifiers over T rounds.
- In each subsequent round, the data points that were misclassified in the previous round are given higher priority.
- AdaBoost is the most popular boosting algorithm.

AdaBoost

- To demonstrate the algorithms, we'll use decision stumps as the weak classifier.
- Decision stumps are decision trees of depth one which classify data points based on just one feature and one threshold.

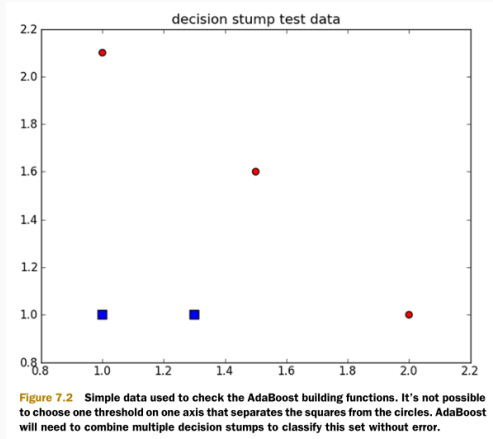


Figure 1: Sample data for decision stumps.

ADABOOST PSEUDOCODE

Given: $(x_1, y_1), \dots, (x_m, y_m)$ where $x_i \in X, y_i \in Y = \{-1, +1\}$

Initialize $D_1(i) = 1/m$.

For $t = 1, \dots, T$:

- Train weak learner using distribution D_t .
- Get weak hypothesis $h_t : X \rightarrow \{-1, +1\}$ with error

$$\epsilon_t = \Pr_{i \sim D_t} [h_t(x_i) \neq y_i].$$

- Choose $\alpha_t = \frac{1}{2} \ln \left(\frac{1 - \epsilon_t}{\epsilon_t} \right)$.
- Update:

$$\begin{aligned} D_{t+1}(i) &= \frac{D_t(i)}{Z_t} \times \begin{cases} e^{-\alpha_t} & \text{if } h_t(x_i) = y_i \\ e^{\alpha_t} & \text{if } h_t(x_i) \neq y_i \end{cases} \\ &= \frac{D_t(i) \exp(-\alpha_t y_i h_t(x_i))}{Z_t} \end{aligned}$$

where Z_t is a normalization factor (chosen so that D_{t+1} will be a distribution).

Output the final hypothesis:

$$H(x) = \text{sign} \left(\sum_{t=1}^T \alpha_t h_t(x) \right).$$

Figure 2: AdaBoost pseudocode [1].

ADABOOST SCHEMATIC

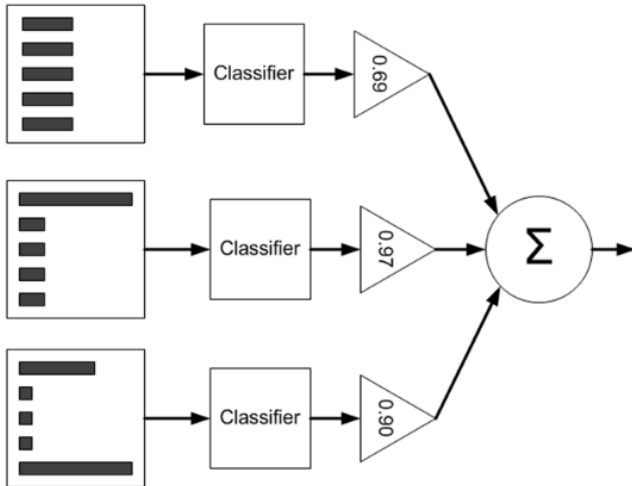


Figure 3: Schematic representation of AdaBoost.

WHY THIS FORMULA FOR α ?

- If α takes the given form and $\alpha > 0$, it can be shown that the classification error exponentially decreases over multiple rounds [2].
- $\alpha_t \geq 0$ if $\epsilon_t \leq 1/2$, which is why we require the weak classifier to have greater than 50% classification accuracy.

CLASS IMBALANCE

WHAT IS IT?

- Let's say we're building a classifier to detect a rare brain tumor from MRI scans.
- In the dataset, for every positive sample there are 100,000 negative samples.
- A model that seeks to minimize classification error will perform poorly at detecting cancer patients.

HOW DO WE DETECT IT?

- Classification error doesn't cut it, we need alternative performance metrics.
- Confusion matrix is useful here.

		redicted	
		+1	-1
Actual	+1	True Positive (TP)	False Negative (FN)
	-1	False Positive (FP)	True Negative (TN)

Figure 4: Confusion matrix for a binary classification problem.

HOW DO WE DETECT IT?

- Precision = $\frac{TP}{TP+FP}$ = fraction of records that were positive from the group that the classifier predicted to be positive.
- Recall = $\frac{TP}{TP+FN}$ = fraction of positive examples the classifier got right.
- Very useful when used together.

HOW DO WE ADDRESS IT?

1. Manipulate the cost matrix.
2. Resample during training.

		Predicted	
		+1	-1
Actual	+1	0	1
	-1	1	0

		Predicted	
		+1	-1
Actual	+1	-5	1
	-1	50	0

Figure 5: Typical (top) and modified (bottom) cost matrices.

REFERENCES I

1. Freund, Y., Schapire, R. & Abe, N. A short introduction to boosting. *Journal-Japanese Society For Artificial Intelligence* 14, 1612 (1999).
2. Freund, Y. & Schapire, R. E. A Decision-Theoretic Generalization of On-Line Learning and an Application to Boosting. *Journal of Computer and System Sciences* 55, 119–139. ISSN: 0022-0000.
<https://www.sciencedirect.com/science/article/pii/S002200009791504X> (1997).

WHY THE NAME?

- Let the training error ϵ_t of h_t be given by $\frac{1}{2} - \gamma_t$.
- Previous learning algorithms required that γ_t be known a priori before boosting begins.
- AdaBoost adapts to the error rates of the individual weak hypotheses, thus the name 'adaptive'.